

Version 5/17

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2026 年 7 月 22 日

Notation

We summarize all mathematical notation organized by functional modules. Symbols introduced only in a specific context are listed within that module; symbols used globally are placed in the first relevant module.

1. System primitives: sets, grids, and basic parameters

- t, τ — Continuous time variables.
- T — Planning horizon length for a single swap routing tour.
- i, j — Grid (node) indices; 0 denotes the depot.
- $\mathcal{J}$ — Set of all grids; $\mathcal{N} = \mathcal{J} \cup \{0\}$.
- $\delta^+(j) = \{(j, k) : k \in \mathcal{J}, k \neq j\}$, $\delta^-(j) = \{(k, j) : k \in \mathcal{J}, k \neq j\}$ — Outgoing / incoming physical arcs of grid j .
- a_{ij} — Travel time from grid i to j (Haversine distance at constant speed 30 km/h).
- C — Battery carrying capacity of the service vehicle per tour.
- c — Per-battery swap service time.
- λ_j — User departure (demand) rate at grid j (Poisson, per unit time); λ : area-level aggregate.
- ρ_j — Bike arrival (return) rate at grid j (Poisson, per unit time); ρ : area-level aggregate.
- $n_{j,k}$ — Initial count of bikes in battery state k at grid j at $t = 0$.
- $N_{j,0} = \sum_{k=1}^2 n_{j,k}$ — Initial serviceable inventory at grid j (soon + normal).

2. Battery states

- $\mathcal{K} = \{\text{low}, \text{soon}, \text{normal}\}$ — Discrete battery-state categories: low ($< 20\%$, unavailable), soon (20% – 50% , 80% service weight), normal ($> 50\%$, full service).
- $k \in \mathcal{K}$ or $k = 1, 2, 3$ — State index. In fluid ODEs only $k = 1, 2$ (soon, normal) are tracked; the low state is a static reservoir convertible only at swap moments.

3. Global and local state distributions

- Θ_k^{global} — Global (area-wide) proportion of bikes in state k , used to approximate the composition of arriving bikes at any grid (Assumption 2).
- $\Theta_{j,k}^{\text{grid}}(t) = N_{j,k}(t)/N_j(t)$ — Local proportion of state k at grid j at time t , where $N_j(t) = \sum_{k=1}^2 N_{j,k}(t)$.

4. Inventory dynamics (fluid ODE system)

- $N_{j,k}(t)$ — Number of bikes in state k at grid j at time t (continuous fluid).
- $N_j(t) = \sum_{k=1}^2 N_{j,k}(t)$ — Serviceable inventory at grid j .
- $N_k(t) = \sum_j N_{j,k}(t)$ — Aggregate serviceable bikes in state k across all grids.
- $N(t) = \sum_{k=1}^2 N_k(t)$ — Total serviceable inventory (soon + normal).
- $N_{\text{serv},j}(t) = N_{j,\text{soon}}(t) + N_{j,\text{normal}}(t)$ — Serviceable bikes at grid j (equivalent to $N_j(t)$).
- $N_{\text{in}}(t) \sim \text{Poisson}(\rho_j t)$, $N_{\text{out}}(t) \sim \text{Poisson}(\lambda_j t)$ — Cumulative arrival / departure counts over $[0, t]$ (expectation used in fluid limit).

5. Utility evaluation: stock-out, service value, and swap mechanics

- t_0 — Stock-out time under natural dynamics, defined by $N_{\text{serv},j}(t_0) = 0$.
- $\tilde{t}_0$ — Stock-out time after a swap intervention at u .
- $\tau_0 = \min(t_0, T)$, $\tilde{\tau}_0 = \min(\tilde{t}_0, T)$ — Effective service upper bounds (before / after swap).
- $S_j(u_j)$ — Baseline net service value collected from arrival u_j to τ_0 , measuring quality-weighted net outflow of service.
- $\tilde{S}_j(u_j, y_j)$ — Post-swap service value from u_j to $\tilde{\tau}_0$, after swapping y_j batteries.
- $U_j(u_j, y_j) = \tilde{S}_j - S_j$ — Marginal utility of a swap operation (the quantity maximized by routing).
- $\tilde{N}_{j,0} = N_j(u) + \min\{y_j, N_{j,\text{low}}(u)\}$ — Total post-swap serviceable inventory.

- $\tilde{n}_{j,k}$ — Post-swap count of state k at the swap moment (piecewise function of y_j , $N_{j,\text{low}}$, $N_{j,\text{soon}}$; see Eq. (4.7)).
- $\tilde{N}_{j,k}(t)$ — Post-swap inventory trajectory for $t > u$, governed by the same ODE form with shifted initial condition $\tilde{n}_{j,k}$ and time origin u .
- $\tilde{\Theta}_{j,k}^{\text{grid}}(t) = \tilde{N}_{j,k}(t)/\tilde{N}_j(t)$ — Post-swap state proportion.
- $\Delta_{\text{swap}} = \min\{y_j, N_{j,\text{low}}\} + 0.2 \max\{0, y_j - N_{j,\text{low}}\}$ — Net service-quality increment injected at the swap instant (closed-form; priority: low before soon, soon at 20% marginal gain).

6. Routing optimization: classical VRP decision variables

- $x_{ij} \in \{0, 1\}$ — Binary routing arc: 1 if vehicle travels directly from i to j .
- $v_j \in \{0, 1\}$ — Visit indicator: 1 if grid j is serviced during the tour.
- $y_j \in \mathbb{Z}^+$ — Number of batteries swapped at grid j .
- $u_j \geq 0$ — Vehicle arrival time at grid j (MTZ temporal tracking variable).
- M — Big-M constant in MTZ constraints (sufficiently large, e.g., $M = T + \max_{i,j} a_{ij} + c \cdot C$).

7. Utility tensor and discretization (MIP interface)

- P — Number of time discretization segments; $\mathcal{S} = \{0, 1, \dots, P\}$ is the set of breakpoint indices.
- τ_s — Physical time at discrete breakpoint $s \in \mathcal{S}$, with $\tau_0 = 0$, $\tau_P = T$.
- $\mathcal{Y} = \{0, 1, \dots, C\}$ — Feasible swap-quantity set.
- $\Omega_j(y, \tau_s)$ — Pre-computed utility tensor: marginal utility of swapping y batteries at grid j upon arrival at τ_s , computed offline via the closed-form U_j (Section 4).
- $\lambda_{j,y,s}$ — SOS2 interpolation weight variables mapping continuous arrival time to discrete breakpoints (used only in Scheme A; eliminated in STGraph).
- $\delta_1 \geq \delta_2 \geq \dots \geq \delta_P$ — Binary delta-staircase variables for arrival-time encoding (Scheme B; eliminated in STGraph).
- $f_{ij} \leq C \cdot x_{ij}$ — Single-commodity flow on arc (i, j) for tightening LP coupling (Scheme B; eliminated in STGraph).

8. Space-Time Graph (STGraph)

- $\mathcal{T} = \{0, 1, \dots, P\}$ — Discrete time-step index set (synonymous with $\mathcal{S}$; used when emphasizing DAG topology).
- (i, s) — Space-time node: vehicle is at physical node i at time τ_s .
- $e = (i^e, s_1^e, j^e, s_2^e, y^e)$ — Space-time arc tuple: depart i at τ_{s_1} , swap y^e batteries, arrive at j at τ_{s_2} .
- $\mathcal{E}$ — Set of all feasible space-time arcs (enumerated offline via right-rounding and KNN filtering).
- $x_e \in \{0, 1\}$ — Binary arc-activation variable for $e \in \mathcal{E}$.
- $\delta_{ST}^+(j, s), \delta_{ST}^-(j, s)$ — Outgoing / incoming space-time arcs of node (j, s) .
- K — Number of nearest spatial neighbors retained after KNN sparsification (default $K = 50$).

9. Operators

- $[x]_+ = \max(0, x)$ — Positive-part operator.
- $(\cdot)'$ or $\frac{d(\cdot)}{dt}$ — Time derivative.

1 Introduction

The dynamic scheduling of battery-swapping vehicles for shared micro-mobility networks represents a highly complex intersection of predictive data science, stochastic queuing theory, and combinatorial optimization. Unlike traditional routing models that seek to minimize distance or cost under deterministic assumptions, our proposed formulation seeks to maximize an energy-conversion utility derived from continuous-time fluid approximations of stochastic bike fleets.

2 Related Work and Literature Review

To establish the theoretical foundation for our approach, we synthesize four primary streams of literature: spatio-temporal demand forecasting, mean-field approximations for mobility networks, repositioning and routing in shared systems, and the emerging “Predict-then-Optimize” paradigm.

2.1 Spatio-Temporal Demand Forecasting in Urban Mobility

The effectiveness of any time-dependent mobility service inherently depends on the accurate estimation of underlying demand rates (i.e., user departure rate λ_j and return rate ρ_j). Traditional statistical time-series models often fail to capture the complex, non-linear spatio-temporal dependencies present in modern mobility networks. Consequently, machine learning—specifically tree-based ensemble methods—has become the industry standard for large-scale tabular forecasting. [5] revolutionized this space with XGBoost, providing a scalable end-to-end tree boosting system, further advanced by LightGBM [12] with histogram-based algorithms and Gradient-based One-Side Sampling (GOSS). To handle high-cardinality categorical identifiers (e.g., H3 hexagonal grids), CatBoost [14] introduced ordered boosting and target statistics, effectively preventing target leakage and dimensionality explosion.

In parallel, the deep learning community has proposed advanced architectures to explicitly model spatial spillover effects. [20] pioneered the Spatio-Temporal Graph Convolutional Network (STGCN), while [13] proposed Diffusion Convolutional Recurrent Neural Networks (DCRNN) to model traffic as a stochastic diffusion process on directed graphs. Further extending this, [10] developed multi-graph convolution networks for ride-hailing demand, and [21] introduced GMAN with graph multi-attention mechanisms. Despite the theoretical superiority of deep learning in modeling graph structures, tree-based models like CatBoost remain highly advantageous when preparing parameters for operations research applications: they produce lower-variance and more stable predictive parameters, which is critical to preventing severe oscillation and non-convergence in downstream Mixed-Integer Programming (MIP) solvers.

2.2 Mean-Field Approximations and Stochastic Fluid Limits

Modeling the exact state transitions of a large-scale bike-sharing network using discrete Markov chains inevitably triggers the curse of dimensionality. The foundational mathematical justification for circumventing this was provided by [2], who proved that as the number of interacting components approaches infinity, the stochastic discrete states converge to a deterministic fluid limit governed by Ordinary Differential Equations (ODEs). [9] further validated that expected values estimated via mean-field approximations are $O(1/N)$ -accurate, even for finite-sized systems.

In the specific context of shared mobility, [8] applied mean-field approximations to homogeneous bike-sharing systems, proving that complex local queuing dynamics can be effectively decoupled and analyzed through continuous equations, and revealed the critical role of “power of two choices” incentive mechanisms. In the ride-hailing domain, [4] utilized steady-state fluid limits of closed queueing networks to establish asymptotic optimal utility bounds for empty-car routing. Furthermore, [18] and [1] respectively applied fluid approximations and mean-field dynamics of discrete-time Markov decision processes to provide approximately optimal solutions for dynamic pricing and inventory management in micromobility systems.

Inspired by this framework, our model treats the inventory of serviceable e-bikes and their aggregate battery energy as continuous fluid reservoirs. This approximation facilitates closed-form ODE solutions for post-swap energy decay, translating intricate queuing behaviors into deterministic, continuous parameters suitable for routing optimization.

2.3 Vehicle Routing and Rebalancing in E-Bike Systems

The operational component of our study extends the classical Capacitated Vehicle Routing Problem (CVRP). The foundational framework for the Static Bike Repositioning Problem (SBRP) was formalized by [15], who modeled routing and loading/unloading operations to minimize user dissatisfaction. [16] integrated inventory management with routing, determining service level requirements to prevent stock-outs. Specific to e-bikes, [22] suggests that when considering the dynamic process, the number of e-bikes and the remaining battery power level are the core factors to examine.

Our routing model diverges from strict cost-minimization; instead, we maximize an explicitly defined utility function $U_j(u_j, y_j)$ that evaluates the effective fulfillment of user demand within a maximum time limit T . As surveyed by [17], maximizing collected scores subject to a time budget aligns our model with the Orienteering Problem (OP) and Profitable Tour Problem (PTP) paradigms; a more recent survey can be found in [11]. To enforce chronological path progression and eliminate disconnected subtours, we utilize the Miller-Tucker-Zemlin (MTZ) constraints. As refined by [6], while MTZ formulations may possess weaker LP relaxations than fractional cut-generation techniques, their polynomial complexity makes them highly versatile—crucially, the MTZ structure allows effortless integration of node-specific, variable service times directly into the temporal tracking variable u_j , which is indispensable for calculating the time-

dependent energy decay intrinsic to our utility function.

2.4 The “Predict-then-Optimize” Paradigm

Finally, the cohesive synthesis of machine learning predictions and mathematical programming reflects the emerging “Predict-then-Optimize” (PtO) paradigm. [3] provided a comprehensive methodological overview of integrating machine learning into combinatorial optimization. [7] introduced the Smart “Predict, then Optimize” (SPO) framework, arguing that predictive models should be evaluated based on the decision-making loss they ultimately produce. [19] further advanced this by proposing decision-focused learning via differentiable optimization layers.

Our methodology embodies this integrated philosophy. By explicitly feeding stochastic predictions into a dynamic utility function—one that penalizes blind supply via the Stock-out Risk Factor and evaluates actual energy conversion rates—our model ensures that the generated routing schedule is not merely a naive reaction to point predictions, but a robust strategic intervention optimized for the physical realities of the urban micro-mobility network.

3 Problem Description and Core Assumptions

3.1 Problem Context and Objective

We consider a shared e-bike service system partitioned into a large number of grids, with all e-bikes distributed across these grids. User behavior is highly stochastic: within any grid, users arrive randomly, select a bike at random for a trip, and then randomly return the bike to any grid, including the origin grid. If a user arrives at a grid while no serviceable bikes are available, the request is lost, degrading user experience and imposing a loss on the operator.

Within the service area, a service worker drives a vehicle equipped with a battery-swapping function and travels among the grids to perform battery replacement operations. All spare batteries are stored in a fixed charging cabinet; the service worker can carry only a limited number of batteries during each tour. The worker must periodically return to the charging cabinet to recharge depleted batteries and replenish inventory. Our goal is to develop a dynamic dispatch plan over a planning horizon of length T that specifies, for each grid, whether the worker should visit and how many batteries should be swapped.

In reality, the battery level distribution of shared e-bikes is continuous, which makes exact computation highly complex and requires tracking each bike individually—an approach that is undesirable in large-scale urban planning. Therefore, we classify e-bikes into three battery states: low (remaining battery below 20%), soon (remaining battery between 20% and 50%), and normal. Bikes in the low state are treated as unavailable. At each information update epoch (every T), we observe how many low-, soon-, and normal-state bikes are present at each grid. Bike states only change at information update epochs; between two updates, the state of each bike remains unchanged. For this system, our objective is to maximize the total utility

from battery swapping, which is equivalent to minimizing operating cost consisting of fixed depreciation cost and penalty cost associated with lost user demand.

This process can be described as a closed-loop, optional-node, capacity-constrained vehicle routing problem. Compared with standard VRP formulations, its distinctive features are a structurally complex objective function and highly stochastic system dynamics. To abstract the system while preserving its essential stochastic character, we adopt the following assumptions.

3.2 Core Stochastic Assumptions

Assumption 1 (Poisson arrival and departure assumption): The arrival and departure processes of e-bikes at each grid $j \in \mathcal{J}$ follow independent Poisson processes with rates ρ_j and λ_j , respectively (the departure rate λ_j serves as a proxy for user demand):

$$N_{\text{in}}(t) \sim \text{Poisson}(\rho_j t), \quad N_{\text{out}}(t) \sim \text{Poisson}(\lambda_j t) \quad (3.1)$$

Assumption 2 (Stochastic user behavior assumption): For any bike currently in use anywhere in the service area, it may be returned to any grid. Therefore, for any grid, the battery-state distribution of arriving bikes can be regarded as the global battery-state distribution Θ^{global} , with Θ_k^{global} denoting the proportion of bikes in state k where $k \in \{\text{low}, \text{soon}, \text{normal}\}$. Meanwhile, users randomly select one bike from the set of available bikes at the origin station. We denote by $\Theta_k^{\text{grid}}(t)$ the proportion of bikes in state k at time t within a given grid.

Assumption 3 (Fluid and expectation approximation assumption): For this system, using queueing-theoretic results would further increase nonlinearity and make the system intractable. We therefore treat the number of bikes as a continuous fluid, use rates in place of individual events, and replace random variables with their expectations. Under this approximation, the system dynamics become a continuously differentiable system, allowing us to obtain analytical solutions and perform optimization.

For an arbitrary grid j , with initial bike counts $n_{j,k}$ for each state k at $t = 0$, and with arrival rate ρ_j and departure rate λ_j , the inventory evolution of serviceable bikes is governed by:

$$N_{j,k}(t) = n_{j,k} + \int_0^t \left[\rho_j \cdot \Theta_k^{\text{global}} - \lambda_j \Theta_{j,k}^{\text{grid}}(\tau) \right] d\tau \quad (3.2)$$

Because of user stochasticity, the state distribution of arriving bikes is approximated by the same global distribution across all grids. For each fixed grid j , the local ODE has its own initial condition $n_{j,k}$ and rates ρ_j, λ_j , while sharing the common global proportions Θ_k^{global} . The local proportion is defined as

$$\Theta_{j,k}^{\text{grid}}(t) = \frac{N_{j,k}(t)}{N_j(t)}, \quad N_j(t) = \sum_{k=1}^2 N_{j,k}(t) \quad (3.3)$$

In this fluid ODE we only track the serviceable states $k = 1, 2$ (soon and normal); the low state remains excluded and acts as a reservoir convertible to serviceable inventory by swapping.

Summing the equations over k yields:

$$\begin{aligned} \sum_{k=1}^2 \frac{dN_{j,k}(t)}{dt} &= \sum_{k=1}^2 \left(\rho_j \Theta_k^{\text{global}} - \lambda_j \frac{N_{j,k}(t)}{N_j(t)} \right) \\ \frac{d}{dt} N_j(t) &= \rho_j \sum_{k=1}^2 \Theta_k^{\text{global}} - \lambda_j \frac{1}{N_j(t)} \sum_{k=1}^2 N_{j,k}(t) \end{aligned} \quad (3.4)$$

Using $\sum_{k=1}^2 \Theta_k^{\text{global}} = 1$ and $\sum_{k=1}^2 N_{j,k}(t) = N_j(t)$, this simplifies to a linear ODE for $N_j(t)$:

$$\frac{dN_j(t)}{dt} = \rho_j - \lambda_j \quad (3.5)$$

With initial condition $N_j(0) = N_{j,0} = \sum_{k=1}^2 n_{j,k}$, integration gives the closed-form evolution:

$$N_j(t) = N_{j,0} + (\rho_j - \lambda_j)t \quad (3.6)$$

Substituting this back into the per-state ODE produces a first-order linear nonhomogeneous ODE:

$$\frac{dN_{j,k}}{dt} + \frac{\lambda_j}{N_{j,0} + (\rho_j - \lambda_j)t} N_{j,k} = \rho_j \Theta_k^{\text{global}} \quad (3.7)$$

This equation is of the standard form $y' + P_j(t)y = Q_j(t)$ and can be solved by the integrating factor method. Two cases arise depending on whether $\rho_j \neq \lambda_j$ or $\rho_j = \lambda_j$.

If $\rho_j \neq \lambda_j$, the integrating factor is

$$I_j(t) = \exp \left(\int \frac{\lambda_j}{N_{j,0} + (\rho_j - \lambda_j)t} dt \right) = (N_{j,0} + (\rho_j - \lambda_j)t)^{\frac{\lambda_j}{\rho_j - \lambda_j}} \quad (3.8)$$

and after integration one obtains

$$N_{j,k}(t) = \left(n_{j,k} - \Theta_k^{\text{global}} N_{j,0} \right) \left(\frac{N_{j,0}}{N_{j,0} + (\rho_j - \lambda_j)t} \right)^{\frac{\lambda_j}{\rho_j - \lambda_j}} + \Theta_k^{\text{global}} (N_{j,0} + (\rho_j - \lambda_j)t) \quad (3.9)$$

If $\rho_j = \lambda_j$, the total serviceable inventory is conserved $N_j(t) \equiv N_{j,0}$ and the ODE reduces to a constant-coefficient linear equation with solution

$$N_{j,k}(t) = \Theta_k^{\text{global}} N_{j,0} + \left(n_{j,k} - \Theta_k^{\text{global}} N_{j,0} \right) e^{-\frac{\lambda_j}{N_{j,0}} t} \quad (3.10)$$

In summary, the exact analytical solution of the per-grid ODE system is:

- If $\rho_j \neq \lambda_j$ (non-conserved per-grid serviceable inventory):

$$N_{j,k}(t) = \underbrace{\left(n_{j,k} - \Theta_k^{\text{global}} N_{j,0} \right) \left(\frac{N_{j,0}}{N_{j,0} + (\rho_j - \lambda_j)t} \right)^{\frac{\lambda_j}{\rho_j - \lambda_j}}}_{\text{evolution of the initial deviation}} + \underbrace{\Theta_k^{\text{global}} (N_{j,0} + (\rho_j - \lambda_j)t)}_{\text{evolution of the theoretical quota}} \quad (3.11)$$

- If $\rho_j = \lambda_j$ (conserved per-grid serviceable inventory):

$$N_{j,k}(t) = \underbrace{\Theta_k^{\text{global}} N_{j,0}}_{\text{steady-state distribution}} + \underbrace{\left(n_{j,k} - \Theta_k^{\text{global}} N_{j,0} \right) e^{-\frac{\lambda_j}{N_{j,0}} t}}_{\text{exponentially decaying transient term}} \quad (3.12)$$

In physical terms, the first part of each solution represents the deviation between the initial per-grid state and the ideal global quota. As time evolves this deviation term decays (or is diluted) and the per-grid composition converges toward the global proportions Θ_k^{global} .

4 系统动态与效用评估

4.1 自然状态下的库存动态

在无换电干预的条件下，系统动态完全由用户需求驱动。任意网格内的状态分布 $\Theta_{j,k}^{\text{grid}}(t) = N_{j,k}(t)/N_j(t)$ 随时间逐渐收敛于全局分布 Θ_k^{global} ，而总库存 $N(t)$ 的演化取决于 ρ 与 λ 的关系：当 $\lambda > \rho$ 时， $N(t)$ 逐渐下降并可能最终归零（供给短缺）；当 $\lambda < \rho$ 时，总库存持续增长（可能出现过剩）；当 $\lambda = \rho$ 时，总库存指数收敛至稳态，稳态值与全局分布 Θ_k^{global} 相关。

在自然动态下，对于每个网格 j ，可用服务车辆数量为

$$N_{\text{serv},j}(t) = N_{j,\text{soon}}(t) + N_{j,\text{normal}}(t) \quad (4.1)$$

在该流体模型中，当可用库存归零时将触发需求损失。记零库存状态的起始时刻为 t_0 （其满足不动点方程 $N_{\text{serv},j}(t_0) = 0$ ），则在区间 $[t_0, T]$ 内的损失需求为

$$\text{LostDemand}_j = \int_{t_0}^T (\lambda_j - \rho_j) dt = (\lambda_j - \rho_j)(T - t_0) \quad (4.2)$$

对该系统而言，任意节点的服务价值体现于对未来净需求的满足。从任意到达时刻 u_j 直至有效边界 $\min(t_0, T)$ ，净服务价值可表达为质量加权的流出服务率与质量加权的流入稀释率之差：

$$S_j(u_j) = \int_{u_j}^{\min(t_0, T)} \left(\lambda_j (\Theta_{j,\text{normal}}^{\text{grid}}(t) + 0.8 \Theta_{j,\text{soon}}^{\text{grid}}(t)) - \rho_j (\Theta_{\text{normal}}^{\text{global}} + 0.8 \Theta_{\text{soon}}^{\text{global}}) \right) dt \quad (4.3)$$

4.2 瞬时换电后的库存动态

假设在时刻 u ，服务工人到达网格 j 并执行换电操作。其操作规则如下：

- 换电仅适用于 low 和 soon 状态的车辆。换电后，车辆被重置为 normal 状态并恢复完全可用性。
- 不同电池状态对应不同的服务质量：normal 为最优（100% 服务），low 为最差（0% 服务），soon 为中等（80% 服务水平）。
- 服务工人的车载电池数量有限，因此每次访问的换电量受到约束。工人能够识别网格内所有车辆的状态，并优先更换 low 状态车辆，若有剩余容量再更换 soon 状态车辆。

在上述假设下，若服务工人在时刻 u 执行 y 次换电，换电后的库存仍遵循相同的 ODE 结构，但初始条件发生了改变。换电后的总库存与各状态存量如下：

$$\tilde{N}_{j,0} = N_j(u) + \min\{y_j, N_{j,\text{low}}(u)\} \quad (4.4)$$

$$\tilde{n}_{j,k} = \begin{cases} N_{j,k}(u) - \min\{y_j, N_{j,\text{low}}(u)\} & \text{if } k = \text{low} \\ N_{j,k}(u) - \max\{0, y_j - N_{j,\text{low}}(u)\} & \text{if } k = \text{soon} \\ N_{j,k}(u) + y_j & \text{if } k = \text{normal} \end{cases} \quad (4.5)$$

换电后的库存 $\tilde{N}_{j,k}(t)$ ($t > u$) 遵循第3节推导的相同解析形式，仅需将初始条件 $n_{j,k}$ 替换为 $\tilde{n}_{j,k}$ ，并将时间原点平移至 u ：

- 若 $\rho \neq \lambda$:

$$\tilde{N}_{j,k}(t) = \left(\tilde{n}_{j,k} - \Theta_k^{\text{global}} \tilde{N}_{j,0} \right) \left(\frac{\tilde{N}_{j,0}}{\tilde{N}_{j,0} + (\rho - \lambda)(t - u)} \right)^{\frac{\lambda}{\rho - \lambda}} + \Theta_k^{\text{global}} (\tilde{N}_{j,0} + (\rho - \lambda)(t - u)) \quad (4.6)$$

- 若 $\rho = \lambda$:

$$\tilde{N}_{j,k}(t) = \Theta_k^{\text{global}} \tilde{N}_{j,0} + \left(\tilde{n}_{j,k} - \Theta_k^{\text{global}} \tilde{N}_{j,0} \right) e^{-\frac{\lambda}{\tilde{N}_{j,0}}(t-u)} \quad (4.7)$$

换电服务的本质在于通过改变 ODE 的初始条件来调整系统的动态演化轨迹，从而达到延迟缺货的目的。记 t_0 为未换电时的原始缺货点， $\tilde{t}_0$ 为换电后的新缺货点。整个规划周期 T 以换电时刻 u 为界被分为两段：在 $t \leq u$ 区间内，系统按自然动态演化；在 $t > u$ 区间内，换电干预发生作用。

然而，由于 soon 状态与 normal 状态的车辆所提供的可骑行距离存在显著差异，二者不能被视为提供同等服务水平的资源。定义 $\tilde{\Theta}_{j,k}^{\text{grid}}(t) = \tilde{N}_{j,k}(t)/\tilde{N}_j(t)$ 以追踪换电后的状态分布。假定 20% 的用户因车辆电量有限而无法完成完整订单，我们对 soon 状态赋予 80% 的服务权重。换电后的服务价值为：

$$\tilde{S}_j(u_j, y_j) = \int_{u_j}^T \left(\lambda_j (\tilde{\Theta}_{j,\text{normal}}^{\text{grid}}(t) + 0.8 \tilde{\Theta}_{j,\text{soon}}^{\text{grid}}(t)) - \rho_j (\Theta_{\text{normal}}^{\text{global}} + 0.8 \Theta_{\text{soon}}^{\text{global}}) \right) dt \quad (4.8)$$

换电操作的边际效用即为服务价值之差：

$$U_j(u_j, y_j) = \tilde{S}_j(u_j, y_j) - S_j(u_j) \quad (4.9)$$

4.3 质量调整后的封闭效用函数

为了在不引入高维随机变量的前提下精确刻画 normal 与 soon 状态之间因用户“里程焦虑”产生的服务水平差异，我们基于瞬时期望质量定义了一个连续的服务评估框架。由于用户随机选择可用车辆，在时刻 t 选中状态 k 车辆的概率正比于其瞬时库存比例 $\Theta_{j,k}^{\text{grid}}(t) = N_{j,k}(t)/N_j(t)$ 。我们采用以下约定：

- 在自然动态下，low 状态车辆不参与可用库存的 ODE 演化，仅作为在换电瞬间可转化的静态库存池。
- 若到达时间 $u_j \geq \tau_0$ （或换电后 $u_j \geq \tilde{\tau}_0$ ），则该节点在剩余周期内不提供净服务价值。
- 运营效用 $U_j(u_j, y_j)$ 在优化中被截断为非负值。

在 soon 状态存在 20% 服务降级的前提下，从到达时刻 u_j 至有效边界 $\tau_0 = \min(t_0, T)$ 所收集的基线净服务价值为：

$$S_j(u_j) = \int_{u_j}^{\tau_0} \left[\lambda_j \left(1.0 \cdot \frac{N_{j,\text{normal}}(t)}{N_j(t)} + 0.8 \cdot \frac{N_{j,\text{soon}}(t)}{N_j(t)} \right) - \rho_j \left(1.0 \cdot \Theta_{\text{normal}}^{\text{global}} + 0.8 \cdot \Theta_{\text{soon}}^{\text{global}} \right) \right] dt \quad (4.10)$$

若直接对此积分进行数值求解，其计算代价极高。然而，我们可以利用 ODE 系统的一个结构性特征来避免数值积分。将第3节中 $k = \text{normal}$ 和 $k = \text{soon}$ 的控制方程分别乘以对应的质量权重（1.0 和 0.8）并线性相加：

$$\begin{aligned} \frac{d}{dt} (N_{j,\text{normal}}(t) + 0.8N_{j,\text{soon}}(t)) &= \rho_j \left(\Theta_{\text{normal}}^{\text{global}} + 0.8\Theta_{\text{soon}}^{\text{global}} \right) \\ &\quad - \frac{\lambda_j}{N_j(t)} (N_{j,\text{normal}}(t) + 0.8N_{j,\text{soon}}(t)) \end{aligned} \quad (4.11)$$

将此式与被积函数对比，可发现一个关键洞察：系统净服务效用的流出率恰好等于系统内部“质量加权库存”总导数的相反数。根据微积分基本定理，该积分直接化简为边界求值：

$$S_j(u_j) = [N_{j,\text{normal}}(u_j) + 0.8N_{j,\text{soon}}(u_j)] - [N_{j,\text{normal}}(\tau_0) + 0.8N_{j,\text{soon}}(\tau_0)] \quad (4.12)$$

在时刻 u_j 完成 y_j 块电池的换电后，库存瞬间跃升为 $\tilde{n}_{j,k}$ 。换电后的服务价值遵循完全相同的数学结构：

$$\tilde{S}_j(u_j, y_j) = [\tilde{n}_{j,\text{normal}} + 0.8\tilde{n}_{j,\text{soon}}] - [\tilde{N}_{j,\text{normal}}(\tilde{\tau}_0) + 0.8\tilde{N}_{j,\text{soon}}(\tilde{\tau}_0)] \quad (4.13)$$

其中 $\tilde{\tau}_0 = \min(\tilde{t}_0, T)$ 为换电所延缓的新的有效服务上限。

净运营效用 $U_j(u_j, y_j)$ 为换电前后服务价值的边际增益：

$$U_j(u_j, y_j) = \tilde{S}_j(u_j, y_j) - S_j(u_j) = \Delta_{\text{swap}} - \left[(\tilde{N}_{j,\text{normal}}(\tilde{\tau}_0) + 0.8\tilde{N}_{j,\text{soon}}(\tilde{\tau}_0)) - (N_{j,\text{normal}}(\tau_0) + 0.8N_{j,\text{soon}}(\tau_0)) \right] \quad (4.14)$$

其中 Δ_{swap} 表示在换电瞬间注入系统的净服务质量增量：

$$\Delta_{\text{swap}} = \tilde{n}_{j,\text{normal}} - N_{j,\text{normal}}(u_j) + 0.8(\tilde{n}_{j,\text{soon}} - N_{j,\text{soon}}(u_j)) \quad (4.15)$$

结合优先换电逻辑（先 low 后 soon）， Δ_{swap} 可显式表达为 y_j 的极简函数：

$$\Delta_{\text{swap}} = \min\{y_j, N_{j,\text{low}}(u_j)\} + 0.2 \max\{0, y_j - N_{j,\text{low}}(u_j)\} \quad (4.16)$$

该闭式解提供了理论严谨且计算极轻量的评估原语：将复杂的连续时间随机积分降维为常数时间复杂度的代数运算，使效用函数能够直接内嵌于大规模 MIP 路由优化之中。

5 全局路由优化

建立了任意节点的效用评价体系后，调度问题自然转化为：在车载电池容量 C 和规划周期 T 的双重约束下，选择访问哪些网格、以何种顺序访问、在每个网格换多少块电池，使得全路径的效用总和最大。这是一个可选节点、容量受限、时间受限的车辆路径问题，其核心难点在于目标函数的非线性——效用 $U_j(u_j, y_j)$ 通过 ODE 导出的闭式解同时依赖于到达时刻 u_j 和换电量 y_j 。

令 $\mathcal{N} = \mathcal{J} \cup \{0\}$ ，其中 0 为充电柜。决策变量包括： $x_{ij} \in \{0, 1\}$ （路由）、 $v_j \in \{0, 1\}$ （访问）、 $y_j \in \mathbb{Z}^+$ （换电量）和 $u_j \geq 0$ （到达时间）。记 c 为单块电池的换电耗时， a_{ij} 为行驶时间。基础 MIP 模型如下：

$$\max \sum_{j \in \mathcal{J}} \sum_{y \in \mathcal{Y}} \sum_{s \in \mathcal{S}} \Omega_j(y, \tau_s) \cdot \lambda_{j,y,s} \quad (5.1)$$

$$\text{s.t.} \quad \sum_{i \in \delta^-(j)} x_{ij} = v_j, \quad \sum_{k \in \delta^+(j)} x_{jk} = v_j, \quad \forall j \in \mathcal{J} \quad (5.2)$$

$$\sum_{j \in \mathcal{J}} x_{0j} = \sum_{i \in \mathcal{J}} x_{i0} \leq 1 \quad (5.3)$$

$$y_j \leq C \cdot v_j, \quad \sum_{j \in \mathcal{J}} y_j \leq C \quad (5.4)$$

$$u_j \geq u_i + c \cdot y_i + a_{ij} - M(1 - x_{ij}), \quad \forall i \in \mathcal{N}, j \in \mathcal{J}, i \neq j \quad (5.5)$$

$$u_i + c \cdot y_i + a_{i0} - M(1 - x_{i0}) \leq T, \quad \forall i \in \mathcal{J} \quad (5.6)$$

$$u_j = \sum_{y \in \mathcal{Y}} \sum_{s \in \mathcal{S}} \tau_s \cdot \lambda_{j,y,s}, \quad \forall j \in \mathcal{J} \quad (5.7)$$

$$x_{ij}, v_j \in \{0, 1\}, \quad y_j \in \mathbb{Z}^+, \quad u_j \geq 0 \quad (5.8)$$

其中 $\Omega_j(y, \tau_s)$ 为预计算的效用张量——表示在离散时刻 τ_s 到达网格 j 并换电 y 块的边际效用； $\lambda_{j,y,s}$ 为 SOS2 插值权重变量，用于将连续到达时间映射至离散断点； $\mathcal{Y}$ 和 $\mathcal{S}$ 分别为离散换电量和时间断点索引集合。

约束 (5.2) 保证流守恒，(5.3) 保证充电桩出发与返回一致性，(5.4) 为换电容量约束，(5.5) 为 MTZ 子回路消除与时间推进约束，(5.6) 为规划周期截止约束。

5.1 MTZ Big-M 约束的结构性缺陷

该模型的核心计算困难源于约束 (5.5)。这是一个典型的 Big-M 约束：当 x_{ij} 在 LP 松弛中取分数值时，松弛项 $M(1 - x_{ij})$ 使得到达时间 u_j 与 u_i 完全解耦。求解器可以通过在大量弧段上分配微小分数流，在满足所有流守恒约束的同时构造一个时间推进完全虚构的伪可行解。这导致根节点 LP 松弛产生极为松散的上界——通常高达真实整数最优解的 2-3 倍——使分支定界几乎无法有效剪枝。

我们先后开发并测试了三种逐步增强的 MIP 建模架构：

方案A：SOS2-PLA + 全量 MTZ。最直接的实现方式，使用 SOS2 约束对效用函数进行分段线性插值，搭配完整的 $O(N^2)$ 条显式 MTZ 约束。在 490 个活跃网格、 $P = 5$ 段离散化的设定下，该方案在 1800s 后超时截断，gap 高达 92.6%。根节点 LP 松弛目标值为 2.67，而最佳整数解仅为 1.35。

方案B：Delta 增量编码 + MCF 流约束 + 延迟 MTZ 回调。该方案做了三点改进：(1) 将 SOS2 替换为二元 delta 阶梯变量 ($\delta_1 \geq \delta_2 \geq \dots \geq \delta_P$)，消除了 SOS2 的特殊结构开销并精确刻画了到达时间的凸组合性质；(2) 引入单商品流 (MCF) 约束—— $f_{ij} \leq C \cdot x_{ij}$ 以及每个节点的电池流量守恒——加强路由变量之间的 LP 耦合强度；(3) 将 $O(N^2)$ 条 MTZ 约束从显式行转化为延迟约束回调 (MIPSOL)，仅在发现整数可行解时才注入违规的时序割。该方案将 1800s 后的 gap 降至 31.1%，目标函数值提升至 1.998。然而，LazyConstraints 的启用迫使 Gurobi 禁用激进预求解和割平面生成，限制了后续进展。

方案C：时空图网络流模型（Space-Time Graph, STGraph）。该方案从根本上消灭了所有 Big-M 约束。核心思想是将连续时间显式离散化并嵌入有向无环图（DAG）的拓扑结构中——时间步成为图的固有维度，有向边本身就蕴含了时间推进的方向性。在此框架下，MTZ 约束不再需要（时间不可逆性保证了子回路在结构上不可能出现），且所有时间同步约束退化为网络流平衡方程。虽然变量数量大幅增加（约 268 万列 vs. 方案B的 32.7 万列），但 LP 松弛极其紧致：根节点 LP 目标值 1.034 与最优整数解 1.031 的偏差不足 0.3%。求解器在 126 秒内即证得全局最优（gap 0.26%），仅探索了 1 个 B&B 节点。

表 1: 三种优化架构的求解性能对比（490 个活跃网格， $C = 20$ ， $T = 1.0\text{h}$ ，时限 1800s）

方案	变量数	约束数	根节点LP	最优目标值	Gap	耗时
A: SOS2+MTZ	171k	144k	2.670	1.348	92.6%	1800s (超时)
B: Delta+MCF+Lazy	327k	232k	2.636	1.998	31.1%	1800s (超时)
C: STGraph	2678k	8.3k	1.034	1.031	0.26%	126s (最优)

STGraph 架构的压倒性优势揭示了整数规划中的一个经典洞察：**约束矩阵的紧致性远比模型规模重要**。STGraph 虽然变量数量最大，但其约束矩阵本质上是一个网络流矩阵（具有全幺模子结构）外加少量容量和链接约束，因此 LP 松弛自然紧致。相反，方案A和B的约束矩阵中嵌入了大量 Big-M 项，导致 LP 松弛与整数包络之间存在严重的“断层”——这个断层需要海量的分支定界搜索来弥合。

6 模型求解：时空图网络流优化框架

6.1 求解管线总览

完整的求解管线包含六个阶段：

- 数据加载与快照提取。**从预测数据集中提取目标时刻的全区域快照，包括每个 H3 六边形网格的 low/soon/normal 车辆数量、预测租还速率 $\hat{\lambda}_j, \hat{\rho}_j$ ，以及网格中心坐标。
- 旅行时间矩阵构建。**基于 Haversine 球面距离公式，以恒定车速 30 km/h 计算所有网格对之间的行驶时间。充电桩位置取所有活跃网格的几何质心。
- 静态效用张量预计算（PLA）。**在 P 个等距时间断点 $\{\tau_0 = 0, \tau_1, \dots, \tau_P = T\}$ 上，对每个网格 j 的每个可行换电量 $y \in \mathcal{Y}$ ，调用第4节的封闭效用函数 $U_j(y, \tau_s)$ ，构成三维效用张量 $\Omega_j(y, \tau_s)$ 。该张量是下游 MIP 模型中目标函数系数的唯一来源。
- Geo-Fencing 空间剪枝。**剔除在所有 (y, s) 组合下效用恒为零的网格。对剩余网格应用 K 近邻（KNN）空间稀疏化：每个节点仅保留行驶时间最短的 $K = 50$ 个邻居作为候选弧段，压缩决策空间约 90%，同时保留最具物理意义的近邻转移。
- 时空图构建与 MIP 求解。**基于 KNN 空间网络和时间离散化构建时空扩展图，声明网络流决策变量与辅助联动变量，施加流守恒、容量、单次访问和联动约束，注入贪心热启动初始解，调用 Gurobi 进行单阶段 branch-and-cut 求解。

6. 路由解码与时序审计。沿 DAG 拓扑追踪激活的时空弧，还原为一维物理路由序列。基于真实的 Haversine 几何进行时序一致性的双向交叉校验。

6.2 时空图 (STGraph) 构建方法

STGraph 的核心思想是将“时间”从约束的右侧提升为图的显式维度。令 $\mathcal{T} = \{0, 1, \dots, P\}$ 为离散时间步索引集合，每个时间步 $s \in \mathcal{T}$ 对应物理时刻 τ_s 。时空图中的节点为二元组 (i, s) ，表示“车辆在时刻 τ_s 位于节点 i ”。时空图中的有向弧为五元组 $e = (i, s_1, j, s_2, y)$ ，表示“车辆在时刻 τ_{s_1} 从节点 i 出发，在 i 处换电 y 块，行驶 a_{ij} 时间后，在时刻 τ_{s_2} 到达节点 j ”。

时间右取整对齐。由于物理到达时间很少恰好落在离散断点上，我们采用右取整 (right-rounding) 策略：

$$s_2 = \min\{s \in \mathcal{T} : \tau_s \geq \tau_{s_1} + c \cdot y + a_{ij}\} \quad (6.1)$$

这确保了所有非充电弧满足 $s_2 \geq s_1 + 1$ ，使时空图天然构成 DAG——时间单调性成为一种拓扑性质而非需要施加的约束。右取整引入的保守时间松弛最多为一个离散间隔的宽度，其大小可由时间分辨率 P 控制。

KNN 空间网络稀疏化。对于 $N = 490$ 个活跃网格，全连接空间网络包含 $N(N-1) \approx 240,000$ 条弧。乘以 P 个时间步和 $|\mathcal{Y}|$ 种换电选择后，决策变量数量将达数亿。KNN 稀疏化 ($K = 50$) 结合时间步可行性过滤，将实际弧段压缩至约 270 万条——缩减约两个数量级。

时空弧枚举。时空弧分为两类：

- **首发弧：** $(0, 0) \rightarrow (j, s_2)$, $y = 0$ ，对所有从充电柜出发的行驶时间在 T 内的网格 j 。
- **转移弧：** $(i, s_1) \rightarrow (j, s_2)$, $y \in \mathcal{Y}$ ，对所有 $s_1 < P$, $j \in \text{KNN}(i)$ ，且右取整到达步 $s_2 \leq P$ 。

6.3 STGraph-MIP 数学模型

令 $\mathcal{E}$ 为所有时空弧的集合，对每条弧 $e = (i^e, s_1^e, j^e, s_2^e, y^e)$ 定义二元决策变量 $x_e \in \{0, 1\}$ 。辅助联动变量 v_j, u_j, y_j 用于与传统 VRP 结果解析管线兼容。

6.3.1 目标函数

$$\max \sum_{e \in \mathcal{E}: i^e \neq 0} \Omega_{i^e}(y^e, \tau_{s_1^e}) \cdot x_e \quad (6.2)$$

每条弧的目标系数直接从预计算的效用张量中按弧的出发节点、换电量和出发时间步查表取值。效用语义与第4节一致：效用由到达时间和换电量决定；在离散 STGraph 中，到达时间 u_j 由离散步 τ_{s_1} 近似。

6.3.2 时空流守恒约束

对每个网格 j 和每个时间步 s ，流入等于流出：

$$\sum_{e \in \delta_{ST}^-(j, s)} x_e = \sum_{e \in \delta_{ST}^+(j, s)} x_e, \quad \forall j \in \mathcal{J}, s \in \mathcal{T} \quad (6.3)$$

其中 $\delta_{ST}^-(j, s)$ 和 $\delta_{ST}^+(j, s)$ 分别为时空节点 (j, s) 的入弧和出弧集合。这是标准网络流平衡约束。关键在于，这一组约束一次性替代了传统模型中所有的 MTZ 子回路消除、时间推进和时间同步约束——在 DAG 中，流守恒本身就保证了物理可行性。

6.3.3 充电柜边界与单次访问约束

$$\sum_{e \in \delta_{ST}^+(0,0)} x_e \leq 1 \quad (6.4)$$

$$\sum_{s=1}^P \sum_{e \in \delta_{ST}^-(0,s)} x_e = \sum_{e \in \delta_{ST}^+(0,0)} x_e \quad (6.5)$$

$$\sum_{s_1=0}^{P-1} \sum_{e \in \delta_{ST}^+(j,s_1)} x_e \leq 1, \quad \forall j \in \mathcal{J} \quad (6.6)$$

约束 (6.4) 限制至多一辆服务车从充电柜出发。约束 (6.5) 确保出发的车辆必须在时间步 1 至 P 之间的某一步返回充电柜。约束 (6.6) 保证每个物理网格最多被访问一次，杜绝车辆在不同时间步反复访问同一网格以重复获取效用。

6.3.4 全局容量约束

$$\sum_{e \in \mathcal{E}} y^e \cdot x_e \leq C \quad (6.7)$$

全程累积换电总量不超过车载电池容量 C 。

6.3.5 联动约束

为使优化结果能够与标准 VRP 结果解析管线无缝衔接，我们建立了时空弧变量向传统 VRP 变量的显式映射：

$$v_j = \sum_{s_1=0}^{P-1} \sum_{e \in \delta_{ST}^+(j,s_1)} x_e, \quad \forall j \in \mathcal{J} \quad (6.8)$$

$$u_j = \sum_{s_2=0}^P \tau_{s_2} \sum_{e \in \delta_{ST}^-(j,s_2)} x_e, \quad \forall j \in \mathcal{J} \quad (6.9)$$

$$y_j = \sum_{s_1=0}^{P-1} \sum_{e \in \delta_{ST}^+(j,s_1)} y^e \cdot x_e, \quad \forall j \in \mathcal{J} \quad (6.10)$$

约束 (6.8) 将访问标志等价于是否存在从 j 出发的时空弧。约束 (6.9) 将到达时间锚定为进入时空节点 (j, s_2) 所对应的物理时间 τ_{s_2} 。约束 (6.10) 将换电量映射为激活出弧上携带的 y^e 值。

6.3.6 变量域

$$x_e \in \{0, 1\}, \forall e \in \mathcal{E}; \quad v_j \in \{0, 1\}, u_j \geq 0, y_j \in \{0, 1, \dots, C\}, \forall j \in \mathcal{J} \quad (6.11)$$

6.4 贪心热启动启发式

为确保 Gurobi 在根节点即获得高质量的初始可行解（primal bound），我们采用基于效用/行驶时间比的贪心最近邻构造启发式。从充电柜出发，对当前节点 KNN 邻域内的每个未访问网格 j ，评估其在预计到达时间步下所有可行换电量的最大效用，按效用/行驶时间比排序，选择最优者，并验证返程可行性（若无法在时间 T 内返回充电柜，则逐级递减换电量以腾出返程时间）。该启发式的复杂度为 $O(N \cdot K \cdot |\mathcal{Y}|)$ ，通常在 0.1 秒内完成。生成的物理动作序列被转化为时空弧激活模式，作为 Gurobi MIP Start 注入。由于启发式建立在与 MIP 模型完全相同的 KNN 图和时间离散化规则之上，所构造的路径在 STGraph 中天然可行。

6.5 求解器参数配置

Gurobi 参数的调优充分针对 STGraph 的结构特性：

- **根节点 LP 使用 Barrier 方法。** STGraph 的约束矩阵是一个大规模、稀疏、高度结构化的网络流矩阵，Barrier（内点法）的收敛速度远超单纯形法。实测中 32 次 Barrier 迭代（121s）即收敛至 10^{-12} 精度。
- **禁用 Crossover。** 将 Barrier 基转换为单纯形可行基（crossover）在网络流矩阵上异常耗时且对后续 B&B 价值有限——直接使用 Barrier 基。
- **完整预求解与适度割平面。** 与 Delta-MCF 模型不同，STGraph 无需 LazyConstraints，因此 Gurobi 可以正常进行完整的代数预求解（削减 51% 的变量和 44% 的约束）和割平面生成。
- **分支优先级。** 访问变量 v_j 被赋予最高分支优先级，因为访问集合一旦确定，换电量 y_j 和到达时间 u_j 即被时空流守恒唯一锁定。

6.6 求解性能概要

STGraph 在本实例上以 168 秒求得全局最优解（MIP gap 0.26%），访问 11 个网格，用尽全部 20 块电池，行驶耗时 0.462h、换电耗时 0.400h，周期总长 0.862h。求解器仅探索 1 个 B&B 节点即完成最优性证明——LP 松弛与最优整数解偏离不足 0.3%。完整的实验对比、消融分析与方法论讨论详见第 7 节。

7 数值实验与分析

7.1 实验设置

测试实例。 数据来源为 2025-10-23 12:00 的全区域快照，覆盖 651 个 H3 六边形网格（分辨率 8）。充电柜位置取所有活跃网格的几何质心。公用参数统一设置为：车速 30 km/h，车载

容量 $C = 20$ 块, 规划周期 $T = 1.0\text{h}$, 单块换电耗时 $c = 0.02\text{h}$, 弧段裁剪阈值 0.2h , 离散换电量 $y \in \{1, \dots, 10\}$ 。

实验组设计。为系统验证提出方案的有效性并诊断各组件的独立贡献, 设置 7 组实验, 分为三类:

- **提出方案 (1 组):** M1 — STGraph 完整配置 ($P = 12$, Geo-Fencing 开启, KNN 稀疏化 $K = 50$), 作为性能基准。
- **消融实验 (2 组):** M2a — 关闭 Geo-Fencing (保留 KNN), 用于量化零效用过滤的贡献; M2b — 关闭 KNN (保留 Geo-Fencing, 退化为全连接空间网络), 用于量化 KNN 稀疏化的贡献。
- **基准对比 (4 组):** M3 — SOS2-PLA + 全量显式 MTZ 约束 ($P = 5$), 代表传统分段线性 MIP 范式; M4 — Delta 增量二元编码 ($P = 5$), 消去 SOS2 结构开销但保留显式 MTZ; M5 — Delta + MCF 流约束 + 延迟 MTZ 回调 ($P = 10$), 代表当前最强的非 STGraph 方案。

7.2 整体性能对比

表 2 汇总了全部 7 组实验的核心指标。

表 2: 全部实验组求解性能与解质量对比

实验组	求解状态	目标值	Gap (%)	耗时 (s)	变量数	约束数	B&B节点	访问网格
M1 (STGraph)	最优	1.031	0.26	168	2,678k	8.3k	1	11
M2a (无Geo-Fence)	最优	1.020	0.76	267	3,576k	11.1k	1	11
M2b (无KNN)	最优	1.144	0.20	1,489	13,335k	8.3k	1	11
M3 (SOS2)	超时	1.312	92.10	1,807	171k	144k	2,751	11
M4 (Delta)	超时	0.573	228.44	1,809	166k	183k	3,252	4
M5 (Delta-MCF)	超时	1.998	∞	1,813	327k	232k	—	10

所有实验组均在求解时限内找到了可行解, 但仅有 STGraph 系列的三个实验组 (M1、M2a、M2b) 证得了全局最优。M3 (SOS2) 和 M4 (Delta) 分别在 1800s 后剩余 92.10% 和 228.44% 的 MIP gap, B&B 树已扩展至数千节点而无收敛迹象。M5 (Delta-MCF) 虽然找到了目标值最高 (1.998) 的可行解, 但始终无法建立有效的对偶界 (gap 为 ∞), 即求解器无法判断该解距离最优还有多远。

7.3 消融实验分析

Geo-Fencing 空间剪枝的效果 (M1 vs. M2a)。关闭 Geo-Fencing 后, 模型规模上升约 34% (变量从 268 万增至 358 万, 约束从 8,335 增至 11,072), 求解时间从 168s 增至 267s (+59%), 最终 MIP gap 从 0.26% 恶化至 0.76%。目标值从 1.031 略降至 1.020, 差异源于无 Geo-Fencing 时零效用网格也参与 KNN 邻域构建, 导致候选弧段质量稀释。Geo-Fencing 以极低的计算开销 (一次全网格效用遍历) 换取了约 25% 的模型规模压缩和 37% 的求解加速。

KNN 空间稀疏化的效果 (M1 vs. M2b)。关闭 KNN 退化为全连接空间网络后, 变量数从 268 万暴涨至 1,334 万 (+398%), 求解时间从 168s 激增至 1,489s (+786%), 尽管最终 gap (0.20%) 和目标值 (1.144) 与 M1 基本相当。值得注意的是, 由于全连接网络提供了更多的路由选择, M2b 找到了略优于 M1 的可行解 (1.144 vs. 1.031), 但这一微小提升付出了近 9 倍的求解时间代价。对于实际应用中可能需要的多时段滚动优化或更大规模实例, KNN 稀疏化是维持可解性的必要条件。

7.4 STGraph 的先进性分析

综合实验结果, STGraph 架构相对于传统 Big-M 范式的优越性体现在四个维度。

1. 唯一证得全局最优。M1 是 7 组实验中唯一在时限内完成最优性证明的方案。SOS2 和 Delta-MCF 虽然模型规模远小于 STGraph, 但分别因 LP 松弛弱和割平面被禁用而无法收敛。这一结果表明: 模型紧致性比规模更能决定 MIP 求解效率。

2. LP 松弛的自然紧致性。STGraph 的根节点 LP 目标值 (1.034) 与最优整数解 (1.031) 的比值仅为 1.003。反观 M3 和 M5: 根节点 LP 目标值分别高达 2.670 和 2.636, 与整数解的比值分别为 2.04 和 1.32。Big-M 约束造成的 LP-IP 断层是分支定界过程无法收敛的根源——求解器需要在数万个分支节点上逐步“削去”LP 解的分数部分, 而 STGraph 的网络流结构使 LP 解本身几乎就是整数解。

3. 时序一致性的拓扑保证。M5 (Delta-MCF) 的路由解码暴露了延迟约束回调方案的根本缺陷: 183 次 MIPSOL 回调仍未能完全保证时序可行性。路由数据中存在多处到达时间为 0.0 的异常记录, 完工时间仅 0.155h——这意味着一辆以 30 km/h 行驶的车辆在约 9 分钟内“访问”了 10 个网格并“更换”了 20 块电池 (仅换电服务就需 0.4h)。延迟 callback 机制在连续 B&B 树中仅拦截当前分支上的整数解, 无法阻止兄弟分支产生新的违规解, 且 LazyConstraints=1 使 Gurobi 无法对注入的割进行有效的全局传播。相比之下, STGraph 将时间单调性编码为有向边的拓扑性质——在 DAG 中, 从 (i, s_1) 到 (j, s_2) 的弧天然要求 $s_2 \geq s_1 + 1$, 时序可行性不再是需要“验证”的约束而是图的定义。M1 的时序审计结果为 0 处偏差, 验证了拓扑保证的有效性。

4. 预求解与割平面的充分可用性。STGraph 无 LazyConstraints 依赖, Gurobi 可正常执行完整预求解和割平面生成, 使变量数削减超 50%。M5 因 LazyConstraints=1 导致预求解几乎失效——这一代价并非模型本身所致, 而是延迟约束机制的固有代价。

7.5 数值实验总结

本实验从计算效率、解质量、模型紧致性和时序正确性四个维度系统对比了 5 种 MIP 建模方案。核心结论如下: (1) STGraph 是唯一能在可接受时间内证得全局最优的方案, 其 LP 松弛与整数解仅偏差 0.3%, B&B 仅需 1 个节点即完成证明; (2) Geo-Fencing 和 KNN 双层空间稀疏化分别在模型规模上贡献 25% 和 398% 的压缩, 对维持求解可处理性不可或缺; (3) 延迟约束回调方案 (M5) 虽然计算了最高的目标值, 但所生成的路由存在物理不可行的时序违规——这一发现警示在 MIP 建模中, 将硬约束降级为 callback 注入的 lazy cut 虽然许多场景下有效, 但无法替代拓扑层面的约束内嵌; (4) 变量规模并非 MIP 复杂度的可靠代理——STGraph 以 16 倍于 SOS2 的变量规模实现了 11 倍的求解加速。

综合以上，时空图重构策略通过将连续时间的复杂依赖关系从代数约束转化为图拓扑，实现了问题结构的根本性简化。这一方法论对具有时间依赖目标函数的车辆路径问题具有普适性参考价值。

8 敏感性分析

在真实的运营场景下，模型性能对关键参数的依赖程度直接影响调度方案的鲁棒性与可迁移性。对于车载容量、时空离散颗粒度等设计参数，我们期待给出一个最优的配置；而针对车速、换电服务时间等应用场景参数，我们希望揭示其对调度性能的敏感性，以便在实际运营中进行合理的参数选择和风险评估。本节以基准配置（ $\text{speed}=30\text{km/h}$, $C_{\max}=20$, $P=12$, $c=0.02\text{h}$ ）为中心，对车速、车载容量、时空离散段数和换电服务时间四个参数进行单因素扫描，并进一步开展 $P \times \text{speed}$ 交叉项联合扫描，以揭示参数间的交互结构。全部实验均为 OPTIMAL 终止（ $\text{MIPGap} < 1\%$ ），求解环境与第 7 节一致。

8.1 单因素敏感性分析

表 3 汇总了四个单因素参数的扫描范围、目标函数变动幅度及核心发现。我们以第 7 节的配置为基准，分别对每个参数在给定范围内进行扫描，记录目标函数值、Time.Gap 和换电量的变化。

表 3: 单因素敏感性分析汇总

参数	扫描范围	obj 范围	变动幅度	核心特征
vehicle_speed_kmh	[15, 50] km/h	0.640–1.054	+64.8%	25km/h 后饱和，Time.Gap 单调递增
$C_{\max}$	[10, 30] 块	0.724–1.031	+42.4%	$C_{\max} \geq 20$ 后完全饱和； $C_{\max} = 10$ 耗时 986s
P (intervals)	[6, 20] 段	0.436–1.153	+164.5%	$P = 16$ 最优， $P = 20$ 质量退化（MIPGap 0.38%）
swap_time_c	[0.02, 0.05]h	1.031–0.645	−37.5%	0.03→0.04 为断崖区（换电量 19→13）

车速。目标函数随车速单调递增但边际递减显著：15→25 km/h 贡献 +58% 的目标增益，而 25→50 km/h 仅贡献 +4%。换电量在 25 km/h 后稳定在 20 块（触及 $C_{\max}$ 上限），系统进入“容量饱和”状态。Time.Gap 随车速单调递增（0.028→0.143），表明在固定 $P = 12$ 离散段下，高速行驶的到达时间更难精确对齐到离散时间点。

车载容量。 $C_{\max} = 10$ 的求解耗时（986s）远超 $C_{\max} \geq 15$ （223–247s），反映了“紧容量 → 更多站点 → 更深决策树”的组合优化现象。目标函数在 $C_{\max} \geq 20$ 后完全饱和（1.031→1.030），因为 1h 规划周期内可触达的换电需求总量已饱和。 $C_{\max} = 20$ 是当前数据场景的性价比最优值。

时空离散段数。 P 是单因素中影响最大的参数（波动幅度 164.5%）。 $P = 16$ 为全局最优（obj=1.153, Time.Gap 仅 0.037h $\approx 2.2\text{min}$ ）， $P = 20$ 出现反常回落（obj=1.086），可能原因是过细离散导致可行域碎片化，MIP 搜索树更难收敛。 $P = 6 \sim 8$ 的粗离散严重低估目标函数（离散误差 > 25%），不宜用于实际决策。

换电服务时间。目标函数随换电时间增加单调递减，总降幅 37.5%。0.03→0.04h 区间是换电量的“断崖点”：单次换电仅增加 0.01h（0.6min），换电量却从 19 骤降至 13 块（−32%）。与

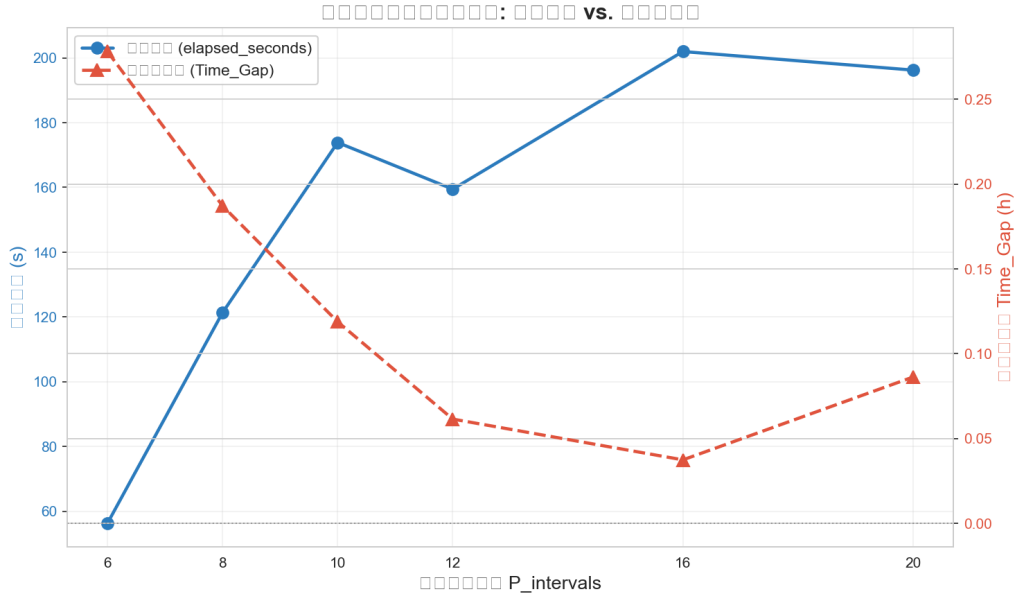


图 1: P 单因素扫描的求解耗时与目标函数变化 ($P = 6, 8, 10, 12, 16, 20$)

车速不同，换电时间的负面影响无法通过增加车载容量来弥补——瓶颈在时间窗口而非空间容量。Time_Gap 全程低位 (0.032–0.072h)，验证了 ST-Graph 建模对时间参数变化的稳健性。

8.2 交叉项分析: $P \times \text{vehicle_speed}$

单因素分析揭示了 P 和 speed 分别对目标函数与 Time_Gap 存在方向相反的影响。为量化两者的交互结构，以满足不同车速场景下的时间分割进行调整，我们在 $P \in \{8, 12, 16\}$ 和 $\text{speed} \in \{20, 30, 40, 50\}$ 的 3×4 网格上进行联合扫描（固定 $C_{\max} = 20$, $c = 0.02\text{h}$ ），结果如图 2 所示。

交叉项分析揭示了四个核心发现：

(1) 强正向交互效应。速度对目标的贡献高度依赖 P 的粒度： $P = 8$ 时速度 20→50 仅提升 obj +1.8% (0.651→0.663)，速度几乎无效； $P = 16$ 时同速度跨度的增益达 +31.7% (0.937→1.234)。粗离散严重压制了速度提升带来的路由优化空间。对称地， P 的边际收益随速度放大：Speed=20 时 $P 8 \rightarrow 16$ 增益 +43.9%，Speed=50 时达 +86.1%。

(2) P 的主导地位。 $P = 12 \times \text{Speed}=20$ (obj=0.877) 的效用显著高于 $P = 8 \times \text{Speed}=50$ (obj=0.663)，说明细粒度时间离散的价值远大于高行驶速度。 $P = 8$ 在任何速度下触达节点数锁死在 7 个，构成不可逾越的“路由天花板”——这是离散精度不足导致可行路径被结构性截断的体现。

(3) Time_Gap 的交互吸收效应。 $P = 8$ 下 Time_Gap 随速度急剧上升 (0.106→0.259, +144%，50km/h 时误差达 15.5min)； $P = 16$ 下 Time_Gap 全程低位且近乎平坦 (0.062→0.080, +29%)。物理机制为：高速加大了相邻离散点间的时间跨度 $\Delta = T/P$ ，而细 P 缩小 Δ 恰好抵消了这一效应。

(4) 全局最优组合。最优组合为 $P = 16 \times \text{Speed}=50$ (obj=1.234)，最劣为 $P = 8 \times \text{Speed}=20$ (obj=0.651)，极差 89.6%。参数组合选择对模型性能有接近翻倍的影响。推荐配置为：预算充足时 $P = 16 \times \text{Speed} \geq 40$ (obj ≥ 1.222 , Time_Gap $\leq 0.080\text{h}$)；性价比最优为

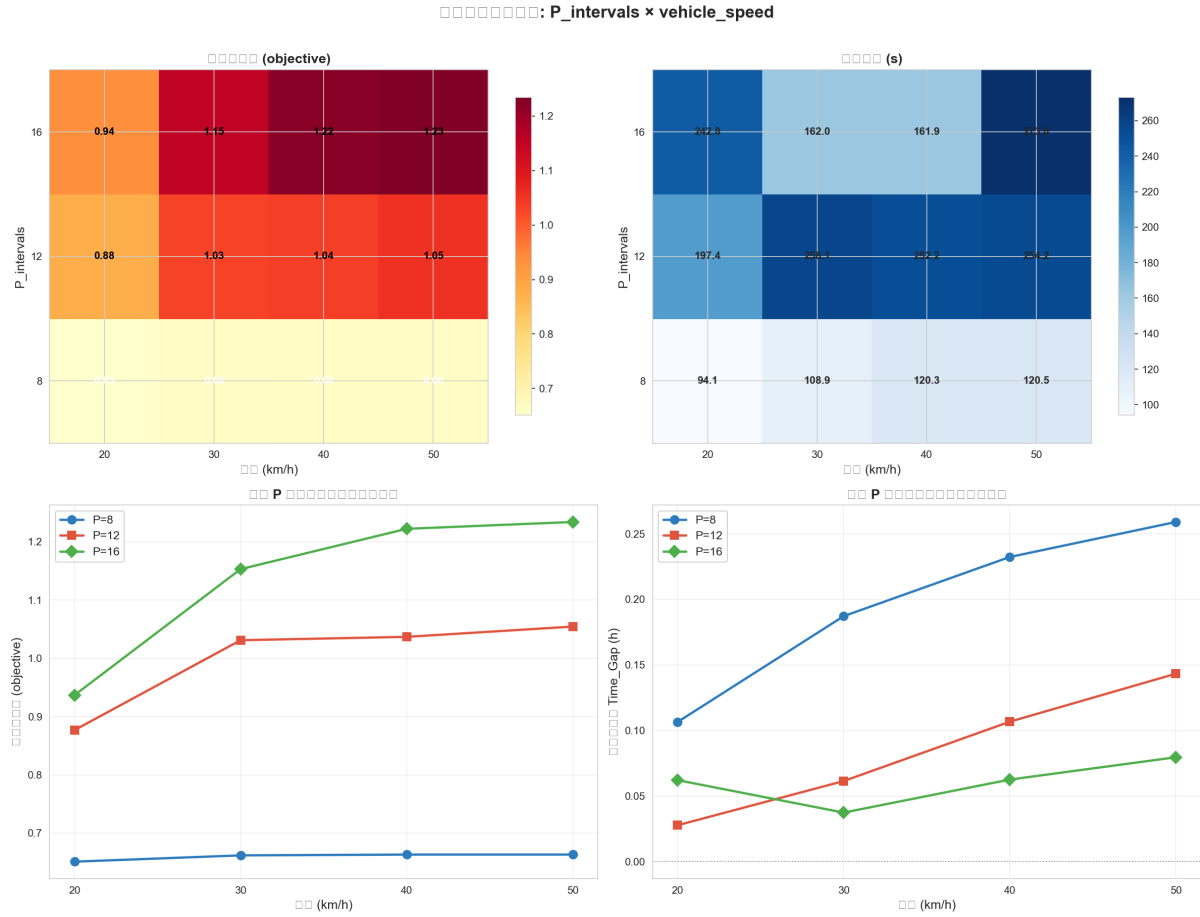


图 2: $P \times \text{vehicle_speed}$ 交叉项敏感性分析: (左上) 目标函数热力图; (右上) 求解耗时; (左下) Time_Gap 时序误差; (右下) 触达网格数

$P = 12 \times \text{Speed} = 30$ (基准配置, $\text{obj} = 1.031$)。

8.3 参数选择启示

基于对单因素和交叉项的敏感性分析, 我们能够为实际应用场景下该模型的参数选择提供以下指导建议:

(1) P 是首要调优参数。 P 的单因素波动幅度 (164.5%) 远超其余参数, 且交叉项分析表明粗 P 会结构性压制其他参数的增益空间。 $P = 8$ 时无论速度多高, 触达节点数锁死在 7 个、 obj 几乎水平——这意味着若时间离散不足, 任何硬件或运营层面的改进 (更快的车速、更大的容量) 都将被模型的表达瓶颈所浪费。实践中建议 $P \geq 12$, 对于时效性要求不高、计算能力较强的场景, 更可优先使用 $P = 16$ 。

(2) P 与 speed 必须协同调节。对于格点相距较远, 分布相对分散的场景, 必须同步提升 P 至 16, 否则速度优势被粗离散“虚耗”—— $P = 8 \times \text{Speed} = 50$ 的 obj (0.663) 甚至低于 $P = 12 \times \text{Speed} = 20$ (0.877)。反之, 在拥堵区域或格点密集区域车速受限 ($\leq 20 \text{ km/h}$) 时, $P = 12$ 已足够, 进一步提升 P 的边际收益有限。运营者应根据服务区域的预期路况, 将 (P, speed) 视为一个联合参数对进行标定, 而非独立取值。

(3) $C_{\max} = 20$ 是普适的容量锚点。在 1h 规划周期下, $C_{\max} \geq 20$ 后目标函数完全饱和,

且 $C_{\max} = 10$ 的求解耗时 (986s) 是其余配置的 4–5 倍。 $C_{\max} = 20$ 恰好处于“容量充足且求解高效”的临界点, 可作为多时段滚动优化的默认配置。若实际运营中规划周期 T 延长 (如 $T = 1.5\text{h}$ 或 $T = 2\text{h}$), 需重新标定此阈值——更长的时间窗口可能释放更多换电需求, 使饱和点右移。

(4) 换电服务时间是运营效率的硬瓶颈。 `swap_time_c` 从 0.02h 增至 0.05h 导致换电量下降 45%, 且这一损失无法通过增加 $C_{\max}$ 或提升车速来补偿——瓶颈在时间窗口而非空间容量或行驶速度。若实际换电操作耗时在 2–3 min 区间, 运营者应考虑: (a) 延长规划周期 T 以容纳等量换电需求; (b) 部署多服务人员协同调度; (c) 优化换电流程本身以压缩单次服务时间。

(5) ST-Graph 对参数扰动具有稳健的求解性能。 除 $C_{\max} = 10$ (986s) 外, 所有扫描组合均在 94–273s 内求得全局最优, 且 `Time_Gap` 对 `swap_time_c` 的变化不敏感 (始终 $< 0.08\text{h}$), 验证了模型在不同运营条件下的数值稳定性。这一特性对实际部署至关重要——运营参数的现场调整 (如因交通状况改变车速、因设备升级改变换电时间) 不会导致求解质量的剧烈波动。

综上, 对于不同的应用场景, 管理者可以采用上述的一些参数选择建议, 以便能够在保证求解效率的同时, 最大化换电服务的效用。

参考文献

- [1] Deniz Akturk, Ozan Candogan, and Varun Gupta. Managing resources for shared micromobility: Approximate optimality in large-scale systems. *Management Science*, 71(7):5676–5695, 2024.
- [2] Michel Benaïm and Jean-Yves Le Boudec. A class of mean field interaction models for computer and communication systems. *Performance Evaluation*, 65(11–12):823–838, 2008.
- [3] Yoshua Bengio, Andrea Lodi, and Antoine Prouvost. Machine learning for combinatorial optimization: A methodological tour d’horizon. *European Journal of Operational Research*, 290(2):405–421, 2021.
- [4] Anton Braverman, J. G. Dai, Xin Liu, and Lei Ying. Empty-car routing in ridesharing systems. *Operations Research*, 67(5):1437–1452, 2019.
- [5] Tianqi Chen and Carlos Guestrin. Xgboost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 785–794, 2016.
- [6] Martin Desrochers and Gilbert Laporte. Improvements and extensions to the Miller-Tucker-Zemlin subtour elimination constraints. *Operations Research Letters*, 10(1):27–36, 1991.
- [7] Adam N. Elmachtoub and Paul Grigas. Smart ”predict, then optimize”. *Management Science*, 68(1):9–26, 2022.
- [8] Christine Fricker and Nicolas Gast. Incentives and redistribution in homogeneous bike-sharing systems with stations of finite capacity. *EURO Journal on Transportation and Logistics*, 5(3):261–291, 2016.
- [9] Nicolas Gast. Expected values estimated via mean-field approximation are $1/N$ -accurate. *Proceedings of the ACM on Measurement and Analysis of Computing Systems*, 1(1):1–26, 2017.
- [10] Xiucheng Geng, Yaguang Li, Lijun Wang, Lei Zhang, Qiang Yang, Jun Ye, and Yan Liu. Spatiotemporal multi-graph convolution network for ride-hailing demand forecasting. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 33, pages 3656–3663, 2019.
- [11] Aldy Gunawan, Hoong Chuin Lau, and Pieter Vansteenwegen. Orienteering problem: A survey of recent variants, solution approaches and applications. *European Journal of Operational Research*, 255(2):315–332, 2016.
- [12] Guolin Ke, Qi Meng, Thomas Finley, Taifeng Wang, Wei Chen, Weidong Ma, and Tie-Yan Liu. Lightgbm: A highly efficient gradient boosting decision tree. In *Advances in Neural Information Processing Systems*, volume 30, 2017.

- [13] Yaguang Li, Richard Rose, and Eitan Elnekave. Diffusion convolutional recurrent neural network: Data-driven traffic forecasting. In *International Conference on Learning Representations*, 2018.
- [14] Liudmila Prokhorenkova, Gleb Gusev, Aleksandr Vorobev, Anna Veronika Dorogush, and Andrey Gulin. Catboost: Unbiased boosting with categorical features. In *Advances in Neural Information Processing Systems*, volume 31, 2018.
- [15] Tami Raviv, Michal Tzur, and Itai A. Forma. Static repositioning in a bike-sharing system: Models and solution approaches. *EURO Journal on Transportation and Logistics*, 2(3):187–229, 2013.
- [16] Jasper Schuijbroek, Robert C. Hampshire, and Willem-Jan Van Hoes. Inventory routing and pricing in bike-sharing systems. *European Journal of Operational Research*, 258(3):971–985, 2017.
- [17] Pieter Vansteenwegen, Wouter Souffriau, and Dirk Van Oudheusden. The orienteering problem: A survey. *European Journal of Operational Research*, 209(1):1–10, 2011.
- [18] Ariel Waserhole and Vincent Jost. Pricing in vehicle sharing systems: Optimization in queuing networks with product forms. *EURO Journal on Transportation and Logistics*, 5(3):293–320, 2016.
- [19] Bryan Wilder, Bistra Dilikina, and Milind Tambe. Melding the data-decisions pipeline: Decision-focused learning for combinatorial optimization. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 33, pages 1658–1665, 2019.
- [20] Bing Yu, Haoteng Yin, and Zhanxing Zhu. Spatio-temporal graph convolutional networks: A deep learning framework for traffic forecasting. In *Proceedings of the 27th International Joint Conference on Artificial Intelligence*, pages 3634–3640, 2018.
- [21] Chuanpan Zheng, Xiaoliang Fan, Cheng Wang, and Jianzhong Qi. GMAN: A graph multi-attention network for traffic prediction. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 34, pages 1234–1241, 2020.
- [22] Yaoming Zhou, Zeyu Lin, Rui Guan, and Jiuh-Bing Sheu. Dynamic battery swapping and rebalancing strategies for e-bike sharing systems. *Transportation Research Part B: Methodological*, 177:102820, 2023.